

ShaTech — ROC, AUC & Softmax Regression Q&A;

1. What is the purpose of the Receiver Operating Characteristic (ROC) curve?

Answer: It evaluates the performance of a binary classifier by plotting TPR vs FPR.

2. What two rates are plotted against each other in the ROC curve?

Answer: True Positive Rate (TPR) and False Positive Rate (FPR).

3. What is the formula for the True Positive Rate (TPR)?

Answer: $TPR = TP / (TP + FN)$.

4. What is the formula for the False Positive Rate (FPR)?

Answer: $FPR = FP / (FP + TN)$.

5. What probability value is varied to trace the ROC curve?

Answer: The classification threshold probability (p^*).

6. What other metrics can be used to understand binary classifier performance?

Answer: Precision and Recall.

7. When threshold probability (p^*) = 1, what is TPR?

Answer: $TPR = 0$, because no positive predictions are made.

8. When threshold probability (p^*) = 1, what is FPR?

Answer: $FPR = 0$, as no negatives are incorrectly classified as positives.

9. When threshold probability (p^*) = 0, what is TPR?

Answer: $TPR = 1$, as all positives are predicted as positive.

10. What are TPR and FPR for a perfect model?

Answer: $TPR = 1$ and $FPR = 0$.

11. What does the ROC curve of a random model look like?

Answer: A diagonal line from (0,0) to (1,1).

12. What can the ROC curve help choose?

Answer: An optimal threshold balancing TPR and FPR.

13. What does AUC stand for?

Answer: Area Under the Curve.

14. AUC-ROC is threshold-independent metric to assess what?

Answer: The classifier's overall discriminative ability.

15. What is AUC-ROC for a perfect model?

Answer: $AUC = 1.0$.

16. What is AUC-ROC for a random model?

Answer: $AUC = 0.5$.

17. What is the typical range of AUC-ROC for real models?

Answer: Between 0.5 and 1.0.

18. What classification involves more than two classes?

Answer: Multi-class classification.

19. What are the two names for regression used in multi-class classification?

Answer: Softmax Regression and Multinomial Logistic Regression.

20. What mathematical function is used for class probabilities $P(y=k|x)$?

Answer: The Softmax function.

21. What does index k denote in softmax probability?

Answer: The class label being considered.

22. What does K represent in categorical cross-entropy loss?

Answer: The total number of classes.

23. How is final predicted value $\hat{y}$ chosen in Softmax Regression?

Answer: By selecting the class with the highest predicted probability.

24. What is the loss function used in Softmax Regression?

Answer: Categorical Cross-Entropy Loss.

25. What is the valid range of probability $P(y=k|x)$?

Answer: Between 0 and 1, inclusive.

26. Application of classification in Mechanical Engineering?

Answer: Fault detection in machinery.

27. Application in Civil Engineering?

Answer: Structural damage prediction.

28. Application in Aerospace Engineering?

Answer: Aircraft fault and anomaly detection.

29. Application in Chemical Engineering?

Answer: Chemical process optimization and safety monitoring.

30. Application in Environmental Engineering?

Answer: Pollution source classification and air quality prediction.